Koji Kanao

Brooklyn, NY |baxin1919@gmail.com

LinkedIn: https://linkedin.com/in/kojikanao GitHub: https://github.com/koji Portfolio: https://koji.thedev.id/

TECHNICAL SKILLS

- Proficient: JavaScript, React.js, Redux, TypeScript, Node.js, Webpack, Python, openFrameworks, Processing, PostgreSQL, Git, Electron, Jest, testing library/react, Storybook, Vite, yarn, Vitest, Jotai, GitHub Actions, Cypress, pnpm, bun, CSS Module, Chakra UI, Cursor, Claude Code
- Exposure: Deno, Gatsbyjs, Nextjs, GLSL, TouchDesigner, OpenCV, Docker, CircleCI, Sketch, rollup.js, three.js, bash, Tailwind CSS

PROFESSIONAL EXPERIENCE

Opentrons | Software Engineer | New York **March 2022 - Present**

- Led UI modernization efforts through a migration to CSS Modules, accessibility improvements, and UI component refactoring.
- Leading AI enablement initiatives by building and maintaining shared harnesses and subagents to automate tasks such as updating npm packages and running linters, accelerating AI adoption and engineering productivity across teams.
- Collaborated with designers to build and launch a UI component library using React and Storybook, which improved component reusability, streamlined code reviews, and increased development speed by 50%
- Led development of an Electron and React-based on-device display application, driving key UX features and enabling cross-team development through a touchscreen-compatible platform.
- Developed and deployed an internal automation tool to automatically post robot usage statistics to Slack, removing repetitive manual tasks and saving roughly 30 minutes per engineer per month
- Delivered the frontend experience for an AI-powered robotics platform showcased at SLAS, leading architecture and implementation of a product widely recognized by conference attendees.
- Added Sentry for production error tracking, resolved critical user-facing issues, and improved the application's error-free session rate to 99.5%
- Reduced CI wait times by 9 minutes from 15 minutes and cut local environment setup time by 80% through migrating from yarn to pnpm

Tayorie | Founding Software Engineer (contract) | Remote **August 2025 -December 2025**

- Built the frontend architecture from scratch for a new LINE application using TypeScript, Next.js, Chakra UI, and Firebase

Pivotal Advisors | Director of Technology | New York **November 2020 - December 2021**

- Rebuilt the company website based on customer data and created internal tools to automate manual business processes, increasing customer inquiries by 15%.

Ubiquiti Networks, Inc | Software Engineer(R&D) | New York **July 2018 - July 2020**

- Led development of a 3D IoT visualization platform using React.js, TypeScript, Three.js, Redux, and PostgreSQL, allowing users to configure and control connected devices directly within browser-rendered digital environments.
- Architected and delivered a reusable npm package that enabled international engineering teams to embed the 3D application into their products with minimal integration effort.
- Optimized the npm package to shorten build time with Webpack, and I could reduce the duration by over 90%
- Built and launched UniFi-Network-AR to simplify IT maintenance tasks using C#, Unity, JavaScript, and Node.js for beta testers
- Developed a machine learning application to detect rooms from an image using Python, Keras, and Beautiful Soup, Selenium, correctly 150-floor plans from Matterport Gallery, processing and predicting 5-floor plans from 3D models

New York University | Researcher | New York **August 2018 - July 2019**

- Provided comprehensive mentorship to students, offering technical guidance on projects and strategic support for their career development and job search
- Developed a low-cost WiFi/Bluetooth signal-sniffing device with Raspberry Pi, Aircrack-ng, Bluez, and Python, contributing to a proof of concept that became a finalist

N-GEMS, Inc | Co-Founder/Lead Engineer | New York **February 2017 - January 2018**

- Built an N-GEMS application using Node.js and Raspberry Pi to capture and visualize signals, and presented a demo-ready prototype, resulting in selection for the NYC Media Lab Combine cohort
- Conducted research on low-cost WiFi/Bluetooth signal sniffing devices using Raspberry Pi, Aircrack-ng, Bluez, and Python, identified the key technologies needed for the N-Gems service, and led the proof of concept. As a result, the team became a finalist in a startup competition and successfully secured small-scale support from investors

SCSK, Inc | System Engineer | Japan **April 2010 - October 2013**

- Developed and maintained power plants' control systems with C, C++, Java, and Unix OS, helping maintain 99% uptime

- Built tools for the project team and manager using VBA, reducing operation time from 5 minutes to 1 minute

EDUCATION

M.A., *New York University - Interactive Telecommunication Program* — May 2018
M.A., *Kobe University - Econometrics* — March 2010
B.S., *Tokyo University of Science - Operations Research* — March 2008

PROJECT WORK

Protocol Factory | Developer | URL — **2026 - present**
A VSCode extension prototype for simulating Opentrons Python protocols. Visualizes Deck state and real-time liquid volume changes via an interactive Webview panel.

Hyouji(表示)| Developer | npm URL | GitHub — **2025 - present**
Hyouji is a simple CLI tool to create/delete labels with GitHub Labels API. Now available as a global npm package with persistent configuration storage

Opentrons Document MCP Server | Developer | npm URL | GitHub — **2025 - present**
A MCP server for Opentrons product such as OT-2, Opentrons Flex, Protocol Designer and Python API